

V4 return analysis

SPY, AAPL, MSFT, and AMZN · four study windows · 1.5-year monthly study

What this report is

- Descriptive statistics of realized equity returns in the same four windows as the 1.5-year monthly 10,000-path study. It does not price options or resimulate paths.
- One table and one four-panel time-series figure per window, always SPY, AAPL, MSFT, and AMZN, always the same five rows and the same return definition.
- The companion notebook is the computational source of truth. This PDF is written from the same payload.

Return definition

- Daily continuously compounded return $R_t = \ln(S_t / S_{t-1})$ from adjusted closes in `prices_clean.csv`.
- Tables: Mean, Std dev. (percent per trading day), Skewness statistic, Ex. kurt. statistic, and Normality statistics (JB), all computed on daily R_t .
- Figures: annualized continuously compounded return $252 \times R_t$, reported in percent.

Tests in the table

- Skewness and excess kurtosis: brackets are p-values from t-tests against the Normal values 0 and 0.
- Jarque-Bera: t-version $JB_t = t_{skew}^2 + t_{kurtosis}^2$ with p-value from $\chi^2(2)$, shown in brackets.

Windows (inclusive)

- Crisis 2008-08-01 → 2009-07-31. Normal 2014-01-01 → 2014-12-31.
- Late-cycle 2018-10-01 → 2019-09-30. COVID 2019-09-01 → 2020-08-31.

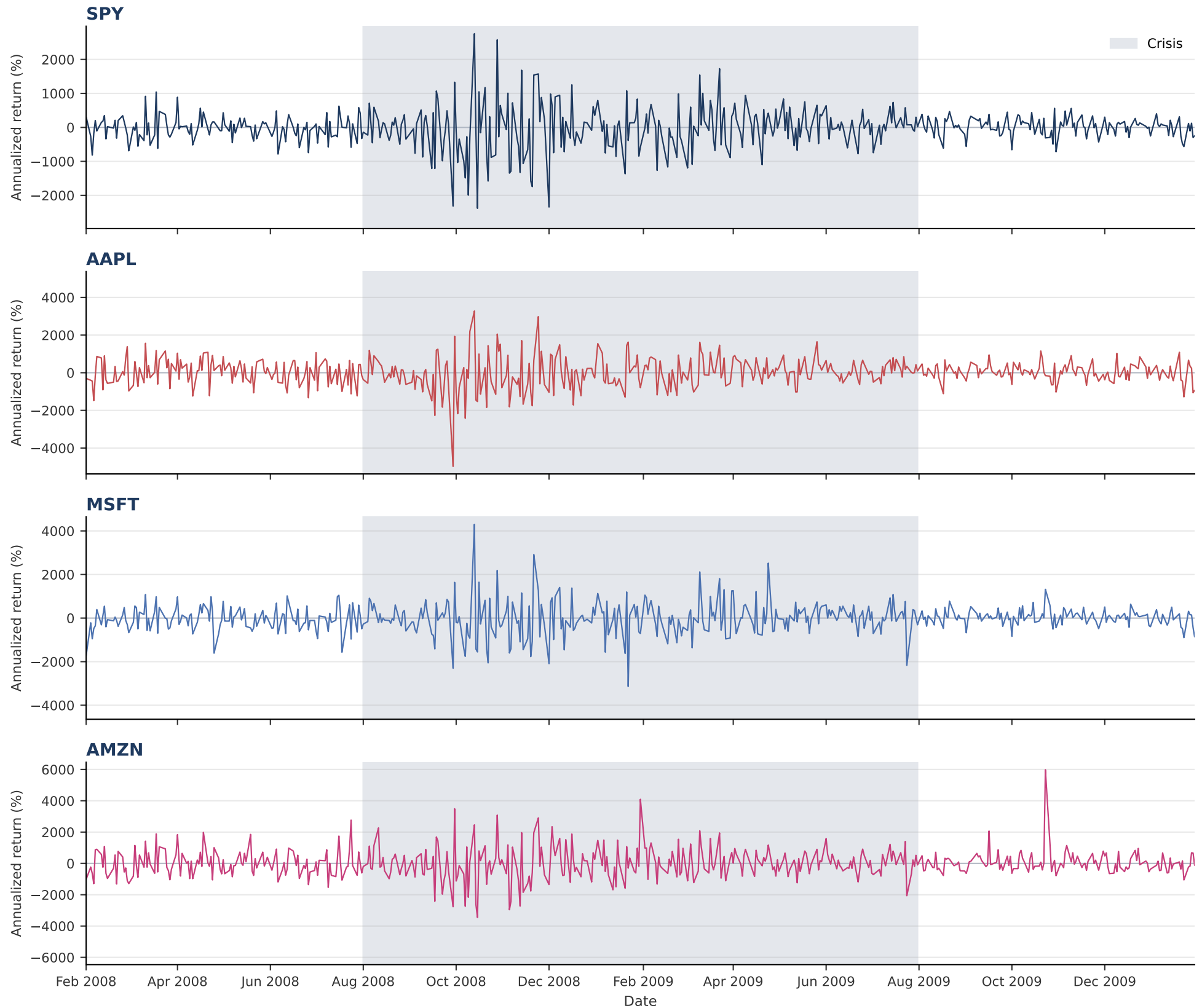
Table R1 · 2008-2009 Crisis · return statistics

2008-08-01 → 2009-07-31 · n = 252 daily observations · R_t in percent per trading day

Statistic	SPY	AAPL	MSFT	AMZN
Mean	-0.0987	0.0109	-0.0260	0.0462
Std dev.	2.8498	3.5650	3.4041	4.2304
Skewness statistic	-0.057 [0.707]	-0.458 [0.004]	0.357 [0.021]	0.135 [0.374]
Ex. kurt. statistic	2.058 [<0.001]	4.157 [<0.001]	3.351 [<0.001]	1.702 [<0.001]
Normality statistics (JB)	15.22 [<0.001]	38.39 [<0.001]	30.05 [<0.001]	12.94 [0.002]

Notes: This table shows sample statistics for the continuously compounded returns, $R_t = \ln(S_t / S_{t-1})$, for SPY, AAPL, MSFT, and AMZN. The sample period is August 1, 2008 to July 31, 2009, for a total of 252 daily observations. Mean return and standard deviation are in percent per trading day. For the skewness and excess kurtosis statistics, the brackets next to the statistics report the p-values from testing the significance of the difference between the empirical values and the theoretical values from the Normal distribution using a t-test. For the normality statistic the p-value of a t-version of the well known Jarque-Bera test for normality is reported in brackets next to the statistics.

Figure R1 · 2008-2009 Crisis · annualized continuously compounded returns



Notes: This figure plots the annualized continuously compounded returns $252 \times R_t$ for SPY, AAPL, MSFT, and AMZN. Date is on the x-axis and the annualized return, in percent, is on the y-axis. Each panel is one name. The shaded band is the evaluation window August 1, 2008 to July 31, 2009 (252 trading days); six months of returns are shown on each side so that band is visible. Large spikes are one-day moves scaled by 252, not cumulative performance.

Table R2 · 2013-2014 Normal · return statistics

2014-01-01 → 2014-12-31 · n = 253 daily observations · R_t in percent per trading day

Statistic	SPY	AAPL	MSFT	AMZN
Mean	0.0425	0.1347	0.0962	-0.0991
Std dev.	0.7165	1.3623	1.1940	2.0636
Skewness statistic	-0.409 [0.009]	-0.366 [0.018]	0.040 [0.789]	-1.810 [<0.001]
Ex. kurt. statistic	1.324 [0.003]	9.459 [<0.001]	1.228 [0.004]	8.441 [<0.001]
Normality statistics (JB)	15.83 [<0.001]	59.68 [<0.001]	8.23 [0.016]	122.53 [<0.001]

Notes: This table shows sample statistics for the continuously compounded returns, $R_t = \ln(S_t / S_{t-1})$, for SPY, AAPL, MSFT, and AMZN. The sample period is January 1, 2014 to December 31, 2014, for a total of 253 daily observations. Mean return and standard deviation are in percent per trading day. For the skewness and excess kurtosis statistics, the brackets next to the statistics report the p-values from testing the significance of the difference between the empirical values and the theoretical values from the Normal distribution using a t-test. For the normality statistic the p-value of a t-version of the well known Jarque-Bera test for normality is reported in brackets next to the statistics.

Figure R2 · 2013-2014 Normal · annualized continuously compounded returns



Notes: This figure plots the annualized continuously compounded returns $252 \times R_t$ for SPY, AAPL, MSFT, and AMZN. Date is on the x-axis and the annualized return, in percent, is on the y-axis. Each panel is one name. The shaded band is the evaluation window January 1, 2014 to December 31, 2014 (253 trading days); six months of returns are shown on each side so that band is visible. Large spikes are one-day moves scaled by 252, not cumulative performance.

Table R3 · 2018-2019 Late-cycle · return statistics

2018-10-01 → 2019-09-30 · n = 251 daily observations · R_t in percent per trading day

Statistic	SPY	AAPL	MSFT	AMZN
Mean	0.0084	0.0030	0.0840	-0.0570
Std dev.	1.0570	2.0348	1.6942	2.2256
Skewness statistic	-0.283 [0.066]	-0.727 [<0.001]	-0.115 [0.450]	-0.059 [0.697]
Ex. kurt. statistic	2.784 [<0.001]	3.702 [<0.001]	1.710 [<0.001]	2.553 [<0.001]
Normality statistics (JB)	24.03 [<0.001]	46.02 [<0.001]	12.74 [0.002]	19.07 [<0.001]

Notes: This table shows sample statistics for the continuously compounded returns, $R_t = \ln(S_t / S_{t-1})$, for SPY, AAPL, MSFT, and AMZN. The sample period is October 1, 2018 to September 30, 2019, for a total of 251 daily observations. Mean return and standard deviation are in percent per trading day. For the skewness and excess kurtosis statistics, the brackets next to the statistics report the p-values from testing the significance of the difference between the empirical values and the theoretical values from the Normal distribution using a t-test. For the normality statistic the p-value of a t-version of the well known Jarque-Bera test for normality is reported in brackets next to the statistics.

Figure R3 · 2018-2019 Late-cycle · annualized continuously compounded returns



Notes: This figure plots the annualized continuously compounded returns $252 \times R_t$ for SPY, AAPL, MSFT, and AMZN. Date is on the x-axis and the annualized return, in percent, is on the y-axis. Each panel is one name. The shaded band is the evaluation window October 1, 2018 to September 30, 2019 (251 trading days); six months of returns are shown on each side so that band is visible. Large spikes are one-day moves scaled by 252, not cumulative performance.

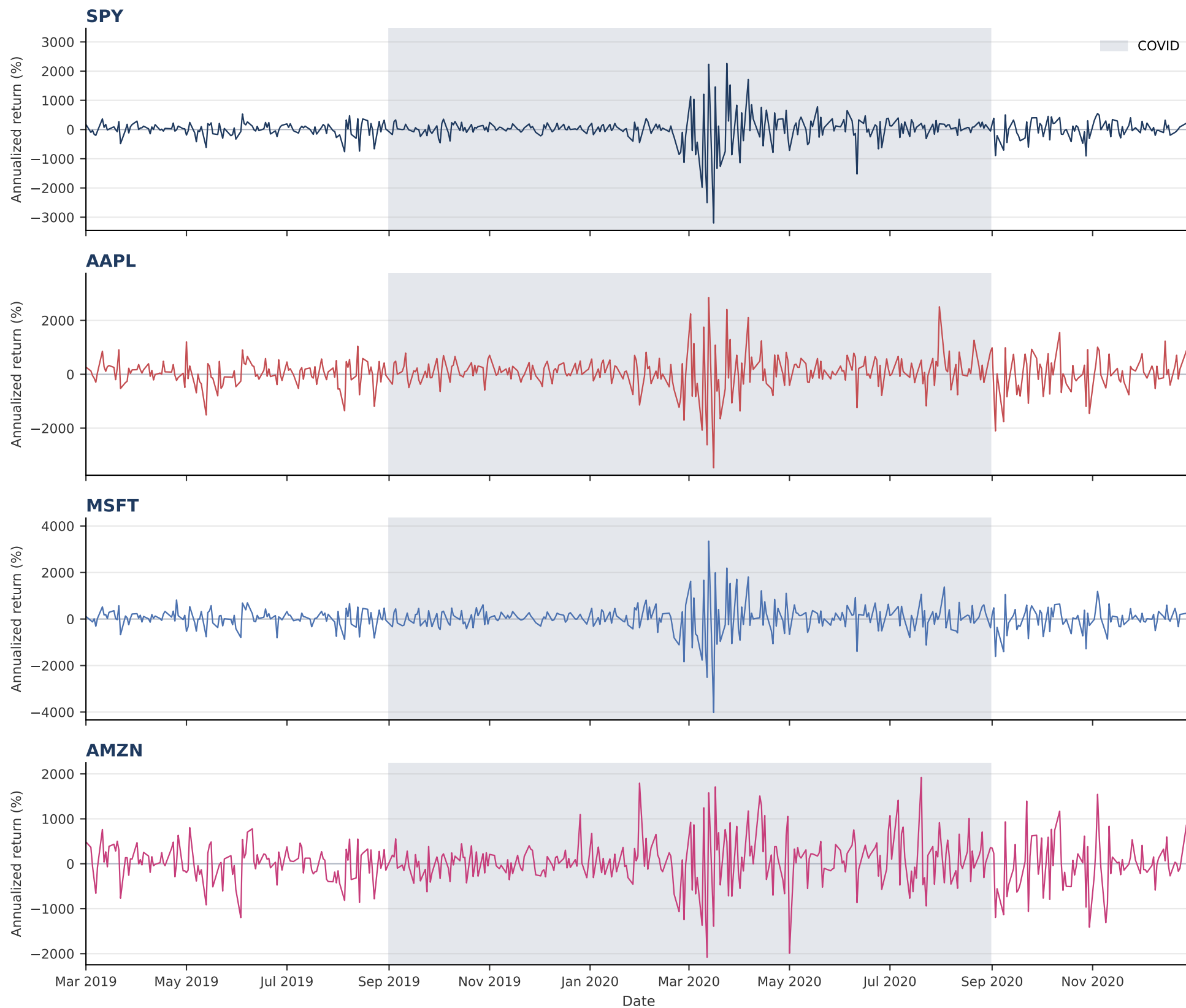
Table R4 · 2019-2020 COVID · return statistics

2019-09-01 → 2020-08-31 · n = 252 daily observations · R_t in percent per trading day

Statistic	SPY	AAPL	MSFT	AMZN
Mean	0.0707	0.3632	0.1998	0.2635
Std dev.	2.1075	2.6341	2.5943	2.1165
Skewness statistic	-0.937 [<0.001]	-0.342 [0.027]	-0.462 [0.003]	0.058 [0.700]
Ex. kurt. statistic	10.321 [<0.001]	6.664 [<0.001]	9.578 [<0.001]	2.861 [<0.001]
Normality statistics (JB)	85.57 [<0.001]	47.96 [<0.001]	62.93 [<0.001]	21.42 [<0.001]

Notes: This table shows sample statistics for the continuously compounded returns, $R_t = \ln(S_t / S_{t-1})$, for SPY, AAPL, MSFT, and AMZN. The sample period is September 1, 2019 to August 31, 2020, for a total of 252 daily observations. Mean return and standard deviation are in percent per trading day. For the skewness and excess kurtosis statistics, the brackets next to the statistics report the p-values from testing the significance of the difference between the empirical values and the theoretical values from the Normal distribution using a t-test. For the normality statistic the p-value of a t-version of the well known Jarque-Bera test for normality is reported in brackets next to the statistics.

Figure R4 · 2019-2020 COVID · annualized continuously compounded returns



Notes: This figure plots the annualized continuously compounded returns $252 \times R_t$ for SPY, AAPL, MSFT, and AMZN. Date is on the x-axis and the annualized return, in percent, is on the y-axis. Each panel is one name. The shaded band is the evaluation window September 1, 2019 to August 31, 2020 (252 trading days); six months of returns are shown on each side so that band is visible. Large spikes are one-day moves scaled by 252, not cumulative performance.